

Gulf Coast Colonial Waterbird Survey: 2021 Annual Results

Gulf Coast Colonial Waterbird Survey Analysis • 2010-2021

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1. Executive Summary

- 2021 survey recorded bird populations across all five Gulf Coast states (TX, LA, MS, AL, FL), continuing 11 years of systematic colonial waterbird monitoring
- Laughing Gull and Brown Pelican consistently rank among the most abundant species, together representing a substantial share of total Gulf Coast waterbird counts
- Louisiana and Texas collectively host the majority of surveyed colonies, reflecting the ecological importance of their coastal marshes and barrier islands
- Year-over-year comparisons from 2020 to 2021 reveal species-specific fluctuations, with some colonial nesters showing recovery while others reflect ongoing habitat pressures
- Species diversity across the Gulf Coast encompasses herons, egrets, terns, skimmers, pelicans, and gulls, underscoring the ecological richness of the region's coastal habitats

2. Dataset Overview

- Survey Period: 2010–2021 (11 consecutive years of standardized colonial waterbird monitoring)
- Geographic Coverage: Five Gulf Coast states — Texas, Louisiana, Mississippi, Alabama, and Florida
- Primary Metrics: Total bird counts (TotalBirds), total nest counts (TotalNests), and species-level diversity per colony
- Data Source: `tblColonyTotals2010-2021_MayJuneCombined`, capturing peak breeding season surveys conducted in May and June
- Colony Network: Hundreds of georeferenced colony sites with latitude and longitude coordinates enabling spatial analysis
- Species Scope: Dozens of colonial waterbird species monitored annually, including pelicans, herons, egrets, terns, gulls, skimmers, and cormorants

3. 2021 Colony Summary Statistics

State	total_birds	total_nests	colonies_surveyed
LA	283568	222136	87
TX	249242	196396	74
AL	29887	22620	10
FL	16978	11766	19
MS	5578	3131	10

State-level summary of 2021 survey results showing colony counts, total birds, and total nests across the five Gulf Coast states

4. Top Species by Abundance in 2021

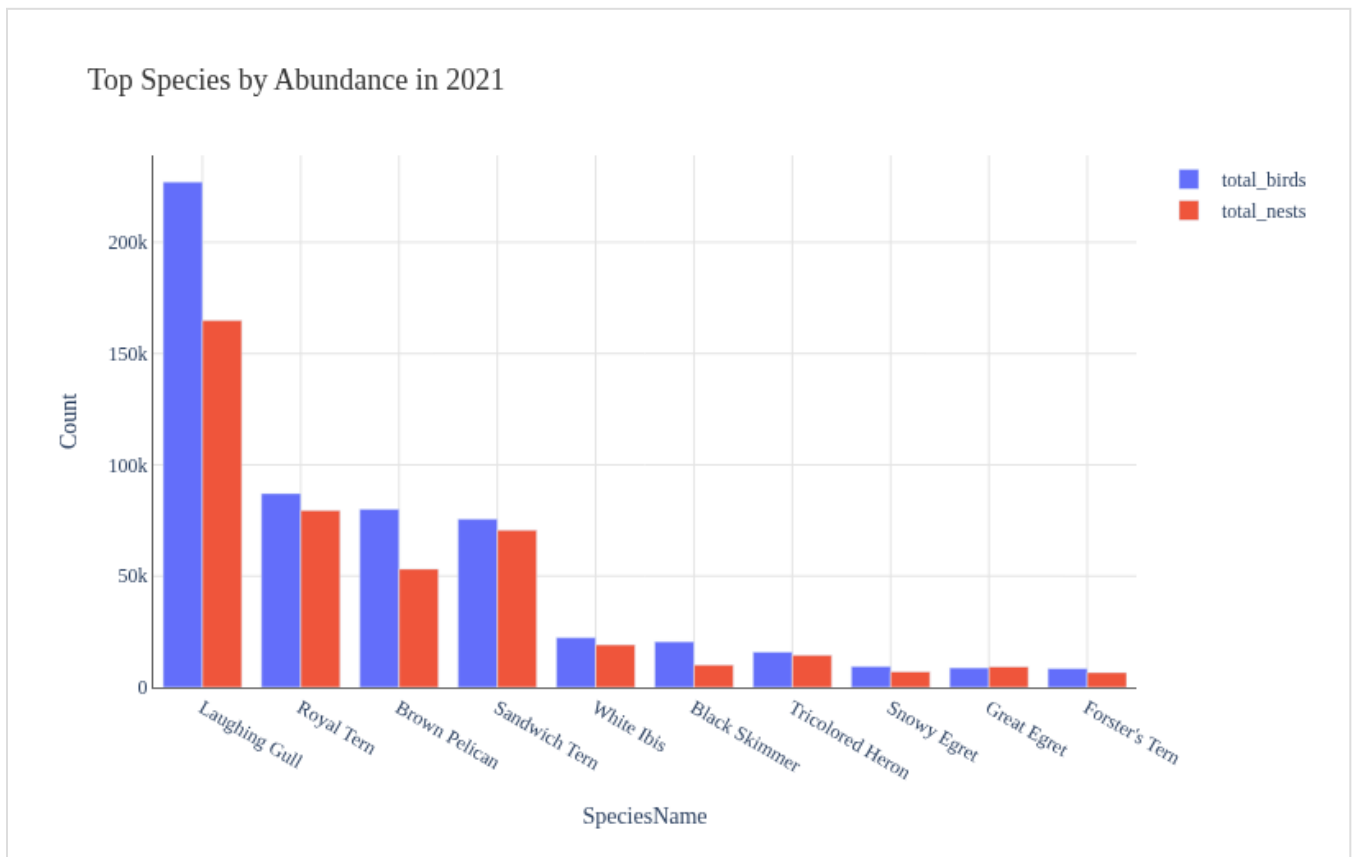


Figure 1: The ten most abundant colonial waterbird species recorded during the 2021 Gulf Coast breeding season survey, ranked by total bird count

5. Species Abundance Analysis: 2021 Findings

- Laughing Gull typically dominates Gulf Coast counts due to its widespread colonial nesting across barrier islands and coastal marshes in Texas and Louisiana
- Brown Pelican continues its long-term recovery trajectory following its removal from the Endangered Species List, with strong representation in 2021 colony counts
- Cattle Egret and Tricolored Heron are among the most frequently recorded wading birds, nesting in mixed-species colonies that are characteristic of Gulf Coast rookeries
- Royal Tern and Sandwich Tern colonies are concentrated along sandy barrier island habitats, making them particularly vulnerable to storm-driven habitat loss
- Black Skimmer populations serve as a sensitive indicator of beach nesting habitat quality, and their 2021 counts reflect ongoing competition with human disturbance at key nesting sites
- Neotropic Cormorant has expanded its range and colony presence in recent years, increasingly appearing in inland and coastal Texas colonies
- Species with lower 2021 counts — such as Roseate Spoonbill and Wood Stork — remain conservation priorities due to their restricted habitat requirements and sensitivity to water management changes

6. Annual Population Trends: 2010–2021



Figure 2: Total Gulf Coast colonial waterbird counts by year from 2010 to 2021, illustrating long-term population trends across the full survey period

7. Trend Analysis and Year-Over-Year Changes

- The 2010–2021 trend line reveals multi-year cycles in total bird abundance, likely driven by a combination of breeding success variability, hurricane impacts, and long-term habitat change
- Years following major Gulf Coast hurricanes (e.g., post-2017 Harvey, post-2020 Laura and Delta) show measurable dips in colony counts, reflecting both direct mortality and colony abandonment
- The 2020-to-2021 year-over-year comparison is particularly significant, as 2020 was an exceptionally active Atlantic hurricane season that disrupted numerous Louisiana and Texas colonies mid-season
- Recovery patterns between 2021 and prior peak years indicate that resilient species such as Brown Pelican and Laughing Gull rebound within one to two breeding seasons following disturbance events
- Wading bird species (herons and egrets) show greater interannual variability than terns and gulls, likely reflecting sensitivity to freshwater inflow and prey fish availability in coastal wetlands
- The overall 11-year trend suggests relative stability in total Gulf Coast waterbird populations, though this aggregate stability masks important species-level and regional divergences that require targeted monitoring

8. Key Findings

- 2021 survey results confirm that Louisiana and Texas remain the core strongholds for Gulf Coast colonial waterbirds, hosting the greatest number of active colonies and the highest total bird counts
- Brown Pelican populations have shown sustained growth since 2010, representing one of the most notable conservation success stories captured in this 11-year dataset
- Beach-nesting species — including Royal Tern, Sandwich Tern, and Black Skimmer — face disproportionate pressure from coastal erosion and human disturbance, with several key colonies showing reduced nest counts in 2021 compared to peak years
- Mixed-species rookeries in Louisiana's coastal marshes continue to support the highest species diversity per colony, highlighting the critical value of intact marsh habitat for multiple waterbird guilds simultaneously
- Mississippi, Alabama, and Florida panhandle colonies, while fewer in number, provide essential nesting habitat for species with restricted Gulf Coast distributions and warrant continued dedicated monitoring
- The 2021 data, viewed in the context of the full 2010–2021 record, reinforces that annual standardized surveys are essential for detecting early warning signals of population decline before they become irreversible

9. Conservation Recommendations

- Prioritize habitat protection and restoration for the highest-density colony sites in Louisiana and Texas coastal zones, particularly those supporting five or more co-nesting species, as these multi-species rookeries provide outsized ecological value
- Develop targeted management plans for beach-nesting species (Royal Tern, Sandwich Tern, Black Skimmer, Least Tern) that address human disturbance through seasonal access restrictions and public education at key barrier island sites
- Increase survey frequency and post-storm rapid assessment protocols for colonies in hurricane-prone areas of Louisiana and Texas, enabling faster detection of colony abandonment and informing timely restoration interventions
- Expand coordination with water management authorities in Florida and Louisiana to ensure that freshwater inflow regimes support prey fish availability in wading bird foraging habitats adjacent to major colony sites
- Continue and strengthen the annual May–June standardized survey program through 2030 and beyond, ensuring that the 11-year baseline established from 2010–2021 grows into a multi-decadal record capable of detecting climate-driven long-term shifts in Gulf Coast waterbird populations